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Winning in Premier League 2: a statistical model of technical performance indicators

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ABSTRACT

Previous research has highlighted the importance of performance indicators in determining match outcomes in men's senior professional football, yet limited evidence exists for elite youth competitions. This study examined the relationship between match outcome (win vs not win) and technical performance indicators in the Premier League 2 and developed a binary logistic regression model to predict match winning. The sample comprised 303 matches from 2022–2024 for model training and 64 matches from 2025 for testing. Mann-Whitney U analyses revealed winning teams executed more forward/backward passes, shots, shots on target, successful crosses, and corners, while facing fewer shots and shots on target ($p < 0.05$). Logistic regression identified increases in shots on target, backward passes, and interceptions increased the probability of winning, whereas shots on target faced, crosses, free kicks, and passes into the final third reduced it. Variables such as successful lateral passes and offensive duels faced were retained in the model to support interpretive context despite non-significance. The final model demonstrated good predictive performance (AUC = 0.771) and generalisability to unseen matches. Findings emphasise the value of creating high-quality scoring opportunities, structured vertical passing, controlled possession, and anticipatory defensive positioning for enhancing success in elite youth football.

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Elite youth football; key performance indicators; match outcome prediction; binary logistic regression; Sports analytics; performance metrics

1. Introduction

Performance analysis (PA) in sport is the systematic evaluation of video and data to inform coaching practices and enhance athletic performance (Groom et al., 2011; Martin et al., 2021; Nelson & Groom, 2012). PA employs tools such as video recording, motion tracking, and statistical software to assess and provide feedback on both individual and team performance (Gabbett, 2016). Analyses are commonly categorised as quantitative, focusing on metrics such as passes, shots, or tackles, or qualitative, examining tactical execution and decision-making (Yi et al., 2019). These approaches may be descriptive, reflecting on past performances, or predictive, aiming to forecast future outcomes (Hughes & Bartlett, 2002). The overarching objective is to

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identify strengths and weaknesses, thereby facilitating informed decision-making, tactical refinement, and skill development. The provision of objective, evidence-based feedback, through PA mitigates the inherent subjectivity of traditional coaching methods (Franks & Goodman, 1986; Nelson & Groom, 2012; Nicholls & Worsfold, 2016) and enables the design of targeted interventions (Davidson et al., 2024). PA further supports strategic planning by allowing coaches to evaluate both their own team and opponents, providing a competitive advantage through evidence-informed tactical preparation (Bauer et al., 2023; Groom et al., 2011; Travis & Butterworth, 2024).

Research in PA has expanded substantially, encompassing multiple interconnected domains that aim to enhance athlete and team performance. One such domain involves the timing and delivery of feedback. Evidence indicates that athletes derive the greatest benefit when feedback is provided at optimal moments, such as prior to key training sessions or competitive fixtures (Groom et al., 2011; Mason et al., 2020). Appropriately timed and individualised feedback has been shown to enhance tactical understanding and technical execution (Travis & Butterworth, 2024; Wright et al., 2016). Match analysis and tactical evaluation constitute additional central components of PA research. Contextual factors, including opposition strengths, match location, and scoreline, influence both tactical decisions and physical output (Groom et al., 2011; Hughes & Bartlett, 2002). These findings are broadly applicable across team sports, where in-game adaptability is critical for success. Technological innovation has also driven substantial evolution in the field. The adoption of GPS, motion capture, and automated video systems has significantly increased the objectivity, accuracy, and granularity of performance data (Gabbett, 2016). These technologies facilitate detailed insights into player behaviour and tactical execution, and their application across sports has promoted methodological cross-fertilisation (Aughey, 2011; Beanland et al., 2014; Heesch & Langdon, 2016). Furthermore, predictive modelling and machine learning approaches have enabled analysts to examine scoring dynamics and statistical trends, facilitating real-time forecasting of match outcomes and tactical responses (Osken & Onay, 2022; Pearson et al., 2019).

A central concept within PA is the identification of key performance indicators (KPIs), which are quantifiable measures used to evaluate the effectiveness of specific actions or behaviours in sport (Herold et al., 2021). KPIs (e.g. scoring metrics, possession data, defensive actions, and physical output measures) provide an empirical basis for monitoring performance, informing tactical or training adjustments, and supporting decision-making (Hughes & Bartlett, 2002). While multiple analytical approaches have been employed to examine KPIs, the general aim is consistent: to identify the performance factors that reliably predict success. In individual sports, regression-based and probabilistic models have been employed to identify performance determinants. For example, O'Donoghue and Cullinane (2011) applied regression analysis to tennis, while Galeano et al. (2022) applied Markov Chains in badminton to quantify the expected probability of winning a rally. In team sports, similar analytical diversity is evident. Higham et al. (2014) applied regression models to rugby sevens to identify performance indicators strongly associated with points scored and match victory. Csataljay et al. (2009) used cluster analysis in basketball to demonstrate that winning teams achieved superior shooting accuracy, free-throw success, and defensive rebounding.

In football, research on KPIs has examined technical, tactical, and psychological dimensions of performance to differentiate successful from unsuccessful teams (Herold et al., 2021; Hughes & Bartlett, 2002). Most studies focus on top European leagues, including England (Jones et al., 2004; Lima et al., 2023; Yang & Kolbinger, 2024; Yi et al., 2019), Spain (Errekagorri et al., 2024; Fernández-Cortés et al., 2024; Yi et al., 2019), and Germany (Jamil et al., 2021; Lepschy et al., 2020; Yi et al., 2019), as well as international tournaments (Branquinho et al., 2024; Farias et al., 2020; Lago-Penas et al., 2011). At senior levels, consistent trends have been identified. Myftiu and Thaqi (2023) analysed the 2022 FIFA World Cup and identified that winning teams produced more shots, higher shot accuracy, and greater shot effectiveness than teams that drew or lost. Corners were positively associated with success, whereas yellow cards, crosses, and conceded corners were negatively associated, likely reflecting defensive vulnerability and ineffective attacking transitions. These findings are further supported by Jamil et al. (2021) and Branquinho et al. (2024), which emphasise the importance of shooting efficiency as a key determinant of match outcomes.

Research by Yi et al. (2019) examining the “Big 5 Leagues” highlighted technical differences in performance data across the Bundesliga (Germany), La Liga (Spain), Ligue 1 (France), Premier League (England) and Serie A (Italy), with significant difference in passing and defending key performance indicators. For example, Bundesliga players made more shots than La Liga, Premier League teams and Serie A teams played more long balls than Ligue 1. Serie A players had less touches of the ball, made less passes and had a lower passing accuracy than players from the other four leagues. Additionally, players from Serie A performed more long passes than Ligue 1 and a lower number of dribbles than players from the Premier League. Stafylidis et al. (2024) observed that winning teams in the Greek Football League generated more goals, shots, and counter-attacks resulting in shots compared with losing teams. Defensive performance has also been considered with Lago-Penas et al. (2011) discovered that losing teams in UEFA Champions League matches accumulated more yellow and red cards, although no significant differences were observed in fouls, corners, or crosses conceded. The exclusion of additional defensive metrics, such as tackles, interceptions, and clearances, represents a knowledge gap with potential to enhance understanding of defensive contributions. Research on possession has produced mixed results with Farias et al. (2020) showing greater ball possession correlates with success, although relationships vary across seasons. Despite substantial research at senior levels, elite youth football particularly within England’s Premier League 2 (PL2), the top under-21 competition, remains underexplored. The PL2 comprises Category 1 academies, featuring players aged 18–21, supplemented by younger prospects and first-team players returning from injury. This diversity results in highly variable match dynamics and team strategies. Although six different clubs have won the league in nine seasons, limited research has examined which KPIs underpin success, leaving a gap in evidence that could inform academy-level tactical development and support the progression of emerging professional players.

A variety of statistical and computational approaches have been applied to investigate the relationships between KPIs and match outcomes, including regression analysis, predictive modelling, and machine learning (Parmar et al., 2017). Linear regression is frequently applied to continuous outcomes such as goals from shot data or league points (Cordon-Carmona et al., 2023) or to assess player market value, identifying predictors

such as age, nationality, and goal contributions (Koglu et al., 2018). For categorical outcomes, logistic regression (dichotomous outcomes) or multinomial models (multiple outcomes) are widely employed. Binary logistic regression is particularly prevalent in football research as it quantifies how variations in performance indicators influence the probability of winning. Although this method does not directly account for margin of victory, some studies have incorporated score differentials to distinguish closely contested from unbalanced games (Gómez et al., 2008; Sampaio & Janeira, 2003; Ziv et al., 2010).

Despite extensive research at senior professional levels, elite youth football, particularly England's Premier League 2 (PL2), remains comparatively underexplored. Unlike senior professional competitions, PL2 represents a hybrid developmental – competitive environment, characterised by age heterogeneity (18–21 years), frequent squad rotation, integration of younger academy players, and occasional inclusion of first-team players returning from injury. These structural characteristics are likely to produce distinct tactical patterns, greater performance variability, and different efficiency profiles compared to senior football. Consequently, it cannot be assumed that performance indicators identified as predictors of success in senior competitions generalise to elite academy contexts. The present study therefore extends KPI-based modelling into the PL2, aiming to (1) investigate differences in performance indicators between winning and non-winning teams, and (2) identify which indicators significantly predict winning using binary logistic regression. By examining performance determinants within a developmental elite setting, this study contributes theoretically by testing the contextual stability of established KPI models, and practically by informing evidence-based coaching strategies within academy football.

2. Methods

2.1. Sample

The overall sample consisted of 367 matches played during 2022–2025 within the Premier League 2 (PL2). The sample was split into a training ($n = 303$ from 2022 to 24) and testing ($n = 64$ from 2025) subsets to ascertain the transferability of the work to future data. Data from the matches was provided by Wyscout (<https://wyscout.hudl.com/app/>), which is a platform for collating match data and video for use within football analysis workflows. Wyscout data itself is collected by professional data analysts who understand both the sport and the technical aspects of data tagging, with 25% of the matches being reviewed in full to ensure the highest accuracy of data possible. According to Pappalardo et al. (2019), the reliability of the Wyscout data assessed using the Kappa statistic was $K = 0.70$, which is considered satisfactory for the analysis and evaluation of performance for professional footballers.

2.2. Data collection and analysis procedure

For the matches, each team had their match outcome (win or not win) and performance data recorded. Match outcome was dichotomised as win versus not win (draw or loss) to reflect the primary focus on identifying determinants of winning. This binary classification

aligns with prior football research employing logistic regression models and allows direct examination of factors that increase the probability of securing victory. While alternative approaches such as multinomial modelling may differentiate draws from losses, the present study prioritised the identification of performance characteristics specifically associated with winning. Twenty-eight (28) performance indicators were selected relating to three groups: offence (shots, shots on target, corners, free kicks, crosses, successful crosses, offensive duels faced, offensive duels won and offsides), defence (shots faced, shots on target faced, defensive duels faced, defensive duels won, aerial duels faced, aerial duels won, interceptions and clearances) and possession (passes, successful passes, forward passes, successful forward passes, backwards passes, successful backwards passes, lateral passes, successful lateral passes, long passes, successful long passes, passes into the final third and successful passes into the final third). All performance indicators and their associated definitions are defined in the Wyscout glossary (<https://dataglossary.wyscout.com>, accessed on 10 February 2025). Clear operational definitions of performance indicators are crucial for several reasons, mainly relating to the consistency, objectivity and accuracy of data collected. Having clear definitions ensures that everyone is on the same page about what is being measured, which reduces bias and allows more precise data to be collected (O'Donoghue, 2014). All indicators were expressed per match to allow standardised comparisons.

2.3. Statistical analysis

Prior to analysis, data were screened for missing or anomalous values. No imputation was required as all variables had complete data. All indicators were expressed per match to standardise comparisons. Descriptive statistics were calculated for the mean (M) and standard deviation (SD). Each performance indicator was assessed for normality (Kolmogorov-Smirnov) and homogeneity of variance (Levene's test), identifying a non-normal distribution ($p < 0.05$). Mann-Whitney U tests were used to investigate differences between technical match performance and match outcome (win vs. not win).

To determine which performance indicators significantly predict winning matches, a binary logistic regression model was constructed using the training dataset. All predictors were initially included in a full model, and multicollinearity was assessed using variance inflation factors (VIF). Highly correlated or redundant predictors were removed to ensure independent predictors, including removal of total passes and subsets with perfect linear dependence. Backward elimination based on AIC was then applied to identify a parsimonious model with only significant predictors retained. Model fit was evaluated using pseudo R^2 measures: Cox and Snell R^2 and Nagelkerke R^2 , as well as -2 Log Likelihood.

The final model was applied to the unseen 2025 test dataset to assess predictive performance. Probabilities of winning were predicted, converted to categorical outcomes using a threshold of 0.5, and compared to the actual match results. Model performance was evaluated using classification accuracy and the area under the receiver operating characteristic curve (AUC), with the ROC curve plotted to visualise discrimination. All analyses were conducted in R (version 4.5.1) using the packages tidyverse, MASS, car, pROC, caret, and rsq.

Table 1. Possession-based metrics and their impact on match outcome (win vs No win).

Variable	Win (<i>n</i> = 246) M ± SD	Draw/Lose (<i>n</i> = 360) M ± SD	U	<i>p</i>
Passes	440.32 ± 108.28	425.93 ± 94.42	41712	0.225
Successful passes	369.52 ± 108.90	352.16 ± 92.22	40942	0.115
Forward passes	149.26 ± 26.11	144.24 ± 24.87	39592	*0.027
Successful forward passes	109.01 ± 26.55	102.66 ± 23.74	38578	*0.007
Backward passes	73.73 ± 21.90	69.18 ± 17.14	39944	*0.040
Successful backward passes	69.90 ± 21.69	65.24 ± 16.87	39738	*0.032
Lateral passes	156.75 ± 54.93	155.29 ± 49.99	44141	0.948
Successful lateral passes	140.59 ± 53.96	138.03 ± 48.30	43597	0.747
Long passes	47.15 ± 10.86	47.65 ± 11.06	45418	0.590
Successful long passes	24.86 ± 7.47	24.72 ± 7.10	43951	0.876
Passes into the final 3 rd	49.06 ± 13.31	49.78 ± 13.04	46673	0.258
Successful passes into the final 3 rd	32.69 ± 11.60	31.99 ± 11.13	43513	0.717

Note: M ± SD – Mean ± Standard Deviation, U – Mann-Whitney U statistic, *p* – *p* value, * – (*p* < 0.05).

3. Results

3.1. Possession metric descriptives

Analysis of possession-based performance indicators revealed varying impacts on match outcome (Table 1). Winning teams had significantly more forward passes (149.26 ± 26.11) than non-winning teams (144.24 ± 24.87; *U* = 39592, *p* = 0.027), and more successful forward passes (109.01 ± 26.55 vs 102.66 ± 23.74; *U* = 38578, *p* = 0.007). Attempted and successful backward passes were also higher for winning teams (73.73 ± 21.90 vs 69.18 ± 17.14, *U* = 39944, *p* = 0.040; 69.90 ± 21.69 vs 65.24 ± 16.87, *U* = 39738, *p* = 0.032). Total passes, successful passes, lateral passes, long passes, and passes into the final third did not differ significantly between outcomes (*p* > 0.05).

3.2. Defensive metric descriptives

Defensive-based performance indicators showed some significant differences between winning and non-winning teams (Table 2). Winning teams faced fewer shots (10.64 ± 4.73 vs 13.60 ± 4.98; *U* = 59582, *p* < 0.001) and fewer shots on target (4.02 ± 2.33 vs 6.43 ± 2.88; *U* = 66468, *p* < 0.001). For duels, only aerial duels won differed significantly, with winning teams having less (12.38 ± 6.30 vs 13.40 ± 6.16; *U* = 49247, *p* = 0.019). Defensive

Table 2. Defensive-based metrics and their impact on match outcome (win vs No win).

Variable	Win (<i>n</i> = 246) M ± SD	Draw/Lose (<i>n</i> = 360) M ± SD	U	<i>p</i>
Shots faced	10.64 ± 4.73	13.60 ± 4.98	59582	*<0.001
Shots on target faced	4.02 ± 2.33	6.43 ± 2.88	66468	*<0.001
Defensive duels faced	83.66 ± 18.73	84.08 ± 18.09	45368	0.607
Defensive duels won	52.54 ± 13.25	52.15 ± 12.37	44257	0.991
Aerial duels faced	27.23 ± 11.31	28.61 ± 11.75	47515	0.126
Aerial duels won	12.38 ± 6.30	13.40 ± 6.16	49247	*0.019
Interceptions	39.43 ± 10.53	38.75 ± 9.89	42652	0.442
Clearances	14.50 ± 7.48	14.23 ± 7.09	44113	0.937

Note: M ± SD – Mean ± Standard Deviation, U – Mann-Whitney U statistic, *p* – *p* value, * – (*p* < 0.05).

Table 3. Attacking-based metrics and their impact on match outcome (win vs No win).

Variable	Win (<i>n</i> = 246) M ± SD	Draw/Lose (<i>n</i> = 360) M ± SD	U	<i>p</i>
Shots	14.22 ± 5.10	11.22 ± 4.76	29746	*<0.001
Shots on target	6.86 ± 2.82	4.18 ± 2.34	20434	*<0.001
Corners	5.48 ± 2.92	4.93 ± 2.69	39732	*0.031
Free kicks	1.83 ± 1.45	2.11 ± 1.61	48776	*0.029
Crosses	13.44 ± 6.02	13.30 ± 6.21	43854	0.840
Successful crosses	4.68 ± 2.73	3.97 ± 2.57	37573	*0.001
Offensive duels faced	83.3 ± 18.00	83.78 ± 18.54	44306	0.990
Offensive duels won	31.61 ± 8.56	31.41 ± 8.91	43456	0.697
Offsides	1.90 ± 1.58	1.86 ± 1.75	42465	0.380

Note: M ± SD – Mean ± Standard Deviation, U – Mann-Whitney U statistic, *p* – *p* value, * - (*p* < 0.05).

duels faced and won, aerial duels faced, interceptions, and clearances did not differ significantly (*p* > 0.05).

3.3. Attacking metrics descriptives

Attacking-based performance indicators revealed several significant differences between winning and non-winning teams (Table 3). Winning teams had more shots (14.22 ± 5.10 vs 11.22 ± 4.76 ; $U = 29746$, $p < 0.001$) and more shots on target (6.86 ± 2.82 vs 4.18 ± 2.34 ; $U = 20434$, $p < 0.001$). Winning teams also had significantly more corners and successful crosses (5.48 ± 2.92 vs 4.93 ± 2.69 ; $U = 39732$, $p = 0.031$; 4.68 ± 2.73 vs 3.97 ± 2.57 ; $U = 37573$, $p = 0.001$) and fewer free kicks (1.83 ± 1.45 vs 2.11 ± 1.61 ; $U = 48776$, $p = 0.029$). Crosses, offensive duels faced and won, and offsides did not differ significantly ($p > 0.05$).

3.4. Binary logistic regression

Prior to the regression analysis, multicollinearity between performance indicators was assessed using Variance Inflation Factors (VIF). Several indicators represented structural subsets of broader variables (e.g. aerial duels won relative to aerial duels faced; successful passes relative to total passes), resulting in inflated VIF values and indicating redundancy. Where strong collinearity was detected, conceptually overlapping variables were evaluated and the less theoretically informative or derivative indicator was removed. This iterative process continued until all retained predictors met acceptable VIF thresholds, ensuring independence of predictors and model stability while preserving variables of substantive tactical relevance. A binary logistic regression with backward selection was then applied to predict Premier League 2 match outcomes as win vs. no-win (Table 4). The model achieved a -2 Log Likelihood of 619.17, with Cox and Snell $R^2 = 0.280$ and Nagelkerke $R^2 = 0.378$, indicating a moderate fit. In summary, the regression showed that backward passes ($B = 0.021$, $p = 0.007$), shots on target ($B = 0.170$, $p < 0.001$), shots on target faced ($B = -0.408$, $p < 0.001$), free kicks ($B = -0.163$, $p = 0.018$), crosses ($B = -0.054$, $p = 0.008$), interceptions ($B = 0.021$, $p = 0.044$), and passes into the final third ($B = -0.025$, $p = 0.013$) were all significant predictors of match outcome. Successful lateral passes ($B = -0.005$, $p = 0.095$) and offensive duels faced ($B = -0.009$, $p = 0.148$) were not statistically

Table 4. Binary logistic regression analysis results for the key predictors of match outcome (win vs No win).

Variable	B	S.E	Wald	ExpB	Lower 95% CI	Upper 95% CI	p
Intercept	0.90	0.827	1.19	2.47	0.49	12.6	0.275
Backward Passes	0.02	0.008	6.89	1.02	1.01	1.04	*0.007
Successful Lateral Passes	−0.01	0.003	2.78	1.00	0.99	1.00	0.095
Passes into the Final Third	−0.03	0.010	6.25	0.98	0.96	1.00	*0.013
Shots on Target	0.17	0.024	50.20	1.19	1.13	1.24	*<0.001
Free Kicks	−0.16	0.069	5.58	0.85	0.74	0.97	*0.018
Shots on Target Faced	−0.41	0.046	78.70	0.67	0.61	0.73	*<0.001
Crosses	−0.05	0.020	7.29	0.95	0.91	0.99	*0.008
Interceptions	0.02	0.011	3.64	1.02	1.00	1.04	*0.044
Offensive Duels Faced	−0.01	0.006	2.25	0.99	0.98	1.00	0.148

Note: * - $p < 0.05$.

significant but were retained in the final model to provide context for related possession and attacking behaviours and to support interpretation of broader team strategies. (Table 4). Cross-validation was conducted to assess the generalisability of the final regression model. A new dataset comprising 64 matches played in 2025 was used, applying the same parameters and the predictors identified in the final model. The model correctly predicted 58.3% of cases, showing only a small drop compared to the training set predictions. The model achieved an area under the ROC curve (AUC) of 0.771 (95% CI: 0.689–0.853; Appendix), indicating moderate to good discriminatory performance.

4. Discussion

The present study aimed to explore the relationship between performance indicators and match outcome in the Premier League 2 (PL2), the most senior and elite youth football league in England. Importantly, PL2 represents a transitional developmental environment rather than simply a lower-quality analogue of senior professional football, and performance patterns observed within this context should therefore be interpreted as developmentally contingent. The primary research question sought to determine whether specific performance indicators could reliably predict match outcome, win versus not win (draw or loss), and thereby enhance the understanding of the determinants of statistical performance at the highest level of academy football in England. By combining descriptive comparisons of possession, attacking, and defensive metrics with a binary logistic regression model, the study provides an integrated perspective linking general performance trends with specific predictors of match outcome.

Descriptively, winning teams executed more forward and backward passes, while lateral passes showed no significant advantage (see Tables 1–3). Defensive metrics also exhibited marked differences with winning teams facing fewer total shots and fewer shots on target, demonstrating the importance of suppressing high-quality opposition attempts (González-Rodenas et al., 2023; Lago-Peñas et al., 2010). Interestingly, non-winning teams won slightly more aerial duels than winning teams (13.40 ± 6.16 vs 12.38 ± 6.30 ; $p = 0.019$). This suggests that dominating aerial contests alone does not necessarily confer a competitive advantage in the PL2, highlighting that effective ball retention and controlled passing may be more critical determinants of success. Other defensive measures,

including interceptions and clearances, did not significantly differ in the descriptive comparisons.

Attacking metrics revealed that winning teams had significantly more total shots and shots on target (Lago-Peñas et al., 2010). Additionally, winning teams produced more successful crosses and corners but committed fewer free kicks. These results suggest that efficient attacking output, particularly in creating and converting high-quality chances, is critical for securing match victories in the PL2. Other attacking indicators, such as total crosses and offensive duels faced or won, did not significantly differ ($p > 0.05$), indicating that the quantity of certain actions may be less important than quality or context.

The binary logistic regression analysis extended the descriptive findings by identifying which indicators independently predicted match outcome when considered simultaneously (Table 4). The final model retained nine predictors, with shooting efficiency and defensive suppression emerging as the most influential factors. The model demonstrated a moderate fit ($-2 \text{ Log Likelihood} = 619.17$; Cox and Snell $R^2 = 0.280$; Nagelkerke $R^2 = 0.378$) and moderate-to-good predictive performance on cross-validation (correct classification = 58.3%; AUC = 0.771, 95% CI = 0.689–0.853; Appendix). While these metrics indicate the model captures meaningful relationships between performance indicators and match outcome, it is important to contextualise them for applied football settings. Moderate accuracy is typical in complex, stochastic environments like football, where outcomes are influenced by many unmeasured factors (e.g. tactics, player decisions, opposition quality). Therefore, the model is better suited for identifying general strategic trends and probabilistic associations rather than predicting individual match outcomes with certainty. Practitioners should interpret the findings as complementary guidance that informs coaching decisions, tactical preparation, and training priorities, rather than as deterministic forecasts for specific games.

Successful lateral passes and offensive duels faced were retained despite non-significance to preserve conceptual completeness within possession and attacking constructs. Their retention enabled more coherent interpretation of passing structure and duel engagement patterns within the broader tactical framework, while maintaining overall model parsimony. This decision reflects an intentional trade-off between strict statistical reduction and the preservation of theoretically meaningful constructs, recognising that overly aggressive variable elimination may compromise interpretability in applied sport contexts. The regression results reinforce several insights from the descriptive analysis while highlighting additional nuances. More shots on target were associated with a higher probability of winning, whereas allowing more shots on target was associated with a lower probability of winning, highlighting that creating high-quality scoring opportunities while suppressing opposition threats generally relates to match success (Anzer & Bauer, 2021; Bandara et al., 2024). Similarly, interceptions were positively associated with winning ($\text{Exp}(B) = 1.02$), suggesting that proactive defensive actions that regain possession can materially influence outcomes. Some possession metrics, such as lateral passes and passes into the final third, were associated with lower probability of winning, suggesting that indiscriminate accumulation of passes may not confer an advantage. Backward passes were associated with higher probability of winning, reflecting their role in maintaining controlled build-up.

Higher numbers of crosses and free kicks were associated with lower probability of winning. This likely reflects reactive match situations, such as teams chasing a deficit, rather than these actions being inherently detrimental. Similarly, non-significant indicators such as successful lateral passes and offensive duels faced were retained. This inclusion provides a conceptual framework for team behaviour, supporting a more holistic interpretation of possession and attacking patterns. These findings highlight that observed associations indicate probabilistic tendencies rather than causal rules, emphasising that match context, tactical decisions, and game state can substantially influence the impact of specific KPIs on outcomes.

Several findings challenge conventional assumptions and previous research. For instance, the positive coefficient for shots on target ($B = 0.170$) contrasts with the findings from some senior-level studies (Harrop & Nevill, 2017), implying that in youth football, an increase in shooting volume does not uniformly translate to improved outcomes. This discrepancy may be due to the quality of the shooting opportunities: better teams are more likely to generate clear-cut chances, whereas lower-quality teams may take speculative or low-probability shots, inflating total shot counts without increasing scoring probability. Similarly, the positive relationship between interceptions and winning contradicts Jamil et al. (2021), who reported decreased success with higher interception rates in youth promotion contexts. In the PL2, this may reflect that successful teams can strategically anticipate opposition play, force turnovers while maintaining possession control. The regression also indicated that increases in crosses and free kicks reduce the likelihood of winning ($\text{Exp}(B) = 0.95$ and 0.85 , respectively). These discrepancies likely reflect the developmental characteristics of PL2, where tactical consistency, physical maturity, and decision-making stability may alter the probabilistic influence of certain KPIs compared with senior professional contexts.

While set pieces are typically scoring opportunities, their effectiveness in PL2 may be limited by lower conversion rates and greater variability in execution, reinforcing that such indicators may operate differently within a developmentally contingent environment. This contrasts with findings from Dimov and Atanasov (2022) and Leite (2020), who report that set-piece success has a stronger probabilistic association with match outcomes at senior levels. The study's findings regarding passing strategy warrant particular attention. Winning teams executed more forward and backward passes, whereas successful lateral passes and passes into the final third negatively predicted match outcome. This suggests that verticality in passing, combined with strategic ball retention, may be more impactful than raw possession metrics. These results resonate with Stafylidis et al. (2024) and Lago-Peñas et al. (2010), who reported that passing structure supports match control but is not a definitive predictor of victory. Additionally, fewer long passes were associated with winning, supporting Mendes et al. (2023) and Warner (2020), who found that shorter, controlled distributions from the back enhance progression and entries into the final third.

Compared with senior professional football, PL2 operates within a structurally distinct performance environment characterised by ongoing physical maturation, greater tactical variability, frequent squad rotation driven by developmental priorities, and an emphasis on progression rather than short-term optimisation. These features likely contribute to the differing behaviour of certain KPIs observed in the present study and underscore why direct generalisation from senior football research should be approached cautiously. For

analysts and coaches working in academy contexts, understanding that these relationships are developmentally contingent is critical for appropriate interpretation and application.

From a practical perspective, these findings have clear implications for coaching in elite youth football. At a general level, coaches can emphasise overarching principles such as shooting efficiency, controlled possession, and proactive defensive engagement, which provide broad guidance on team performance. The data also highlight specific KPIs that indicate when these standards are being met. This allows feedback to be personalised to the team or individual players. For example, prioritising high-quality shooting opportunities rather than volume aligns with the general standard of efficient attacking, while monitoring shots on target or the number of backward passes provides tangible, measurable indicators for both practice and match play.

Training drills that replicate match-like shooting scenarios, incorporating decision-making under pressure, can improve both shot selection and conversion. Ball retention and passing quality should be emphasised, with structured build-up play, rondo exercises, and short passing patterns forming key elements of technical sessions. Defensive training should focus on anticipatory skills and pressing to maximise interceptions often seen within modern football (Brindescu et al., 2021). These exercises allow coaches to link general performance standards with observable, player-level metrics, providing actionable insights that bridge theory and practice.

The present study demonstrates that predictive models can meaningfully complement these traditional coaching practices. By integrating quantitative analyses with observational feedback and qualitative judgement, coaches gain an evidence-based lens to identify probabilistic tendencies and tactical efficiencies. In practice, this means that general principles (“be efficient at retaining possession”) can be translated into targeted interventions using specific KPIs (“shots on target reach X per game”), enabling more structured and informed performance optimisation. The methodological rigour of combining descriptive analysis with binary logistic regression, addressing multicollinearity via backward elimination, and validating the model on an unseen dataset strengthens the reliability and applicability of these insights to future PL2 matches.

In practice, these predictive insights can guide coaching decisions before, during, and after matches. For example, within a typical weekly PL2 microcycle, an analyst might begin Monday’s post-match review by presenting model-derived KPIs (e.g. shots on target, shots on target faced, backward passes) alongside corresponding video clips. If the model indicates that shots on target were below the team’s season average in a loss, video analysis may reveal poor shot selection or rushed decision-making in the final third. This could inform midweek training sessions focused on structured attacking patterns and finishing under pressure. Similarly, if shots on target faced were elevated, analysts might combine defensive transition clips with spatial data to highlight breakdowns in pressing or compactness, shaping tactical adjustments for the upcoming fixture. During the pre-match meeting, coaches could reinforce key probabilistic indicators identified by the model, translating them into simple behavioural cues (e.g. “protect central zones”, “prioritise shot quality over volume”). In this way, statistical outputs are not treated as isolated metrics but are embedded within an integrated workflow that combines quantitative modelling, video feedback, and coach observation. Within PL2’s dual developmental-competitive structure, however, such KPI targets must be balanced against long-term learning objectives,

positional experimentation, and exposure to varied tactical demands, reinforcing that performance optimisation in academy football extends beyond immediate match outcome.

5. Future research and limitations

The binary classification of match outcome may be viewed as oversimplifying competitive reality, as draws are treated equivalently to losses; however, it must be emphasised that the focus here was on match success (i.e. winning). Aggregate match-level data do not account for individual player contributions, tactical systems, or coaching interventions, all of which could significantly influence outcomes. The model's predictive performance, while moderate, may not generalise across leagues or seasons without further validation. Finally, contextual variables such as red cards, injuries, fixture congestion, and home versus away status were not incorporated, limiting the comprehensiveness of the model.

Future research could address these limitations and further expand understanding of youth football performance. Incorporating a multinomial logistic regression with three outcome categories (win, draw, loss) would more accurately reflect the competitive context of league play. Including contextual variables and player-level metrics could improve model precision and interpretability. Longitudinal analyses across multiple seasons would allow assessment of the stability and generalisability of key predictors, while combining correlated indicators (e.g. aerial duels faced and won into a success rate) could reduce multicollinearity and enhance interpretability. More broadly, future models may evolve towards hybrid analytical frameworks that integrate technical KPIs with contextual (e.g. match status, opposition strength) and tactical or spatial variables, reflecting emerging trends in football performance analytics. Comparative studies across youth leagues in different countries would elucidate contextual and cultural factors that influence match outcome determinants, further bridging statistical analysis and applied coaching practice.

From a player recruitment and development perspective, longitudinal analysis could be undertaken on youth players as they progress through the academy system. Here, age-related benchmarks could be retrospectively calculated to provide clear objective criteria to highlight players with future elite potential (Frost et al., 2025). Similarly, such benchmarking have started to be used to identify managers based on style of play or the teams "game model" measured by a range of technical KPIs like those highlighted within the current research (see Bowyer et al., 2025). Therefore, this research approach offers a fruitful avenue to further understand team performance, player development, and both player and manager recruitment.

6. Conclusion

The present study demonstrates that match outcomes in the PL2 are shaped by a combination of shooting efficiency, defensive suppression, and structured possession behaviours. Regression analysis identified shots on target, backward passes, and interceptions as positive predictors of winning, while indicators associated with defensive exposure and reactive attacking phases reduced winning probability. These findings

underscore the importance of tactical efficiency, vertical passing, and controlled possession in achieving success at the elite youth level. Coaches and performance analysts can apply these insights to training design, tactical preparation, and match strategy to optimise team performance in the PL2. It should be emphasised that such quantitative models provide strategic insight rather than match-by-match predictions; their outputs are most valuable when combined with qualitative assessments, video analysis, and contextual understanding of player and team behaviour.

Author contributions

CRedit: **Nick Howell:** Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft; **Ryan Groom:** Conceptualization, Methodology, Writing – review & editing; **Scott B. Nicholls:** Conceptualization, Methodology, Project administration, Supervision, Writing – review & editing.

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Appendix. ROC curve with AUC and 95% CI